

Honours Question Bank IA2 - Answers

Subject: Text, Web and Social Media Analytics

Semester: VIII

SHORT QUESTIONS

Q1. What are the types of data in web Usage Mining?

- Web Usage Mining analyzes user interaction data to understand browsing patterns and behavior. The main types of data are:
- **Server Log Data:** Records stored by web servers containing information about user requests, including IP addresses, timestamps, requested URLs, HTTP methods, status codes, and user agents. This is the primary source for usage mining.
- **Application Server Data:** Data from application-level logs that track user interactions with dynamic web applications, including shopping cart activities, form submissions, and database queries.
- **Application Level Data:** Business event data captured through custom logging mechanisms within web applications, tracking specific user actions like product views, searches, downloads, and transactions.
- **Clickstream Data:** Sequential record of pages visited by users, capturing the navigation path and click patterns through the website.
- **User Registration/Profile Data:** Demographic and personal information provided by users during registration, including age, location, preferences, and interests.

Example: An e-commerce website like Amazon collects server logs showing that a user from IP 192.168.1.1 visited the electronics category at 2:30 PM, viewed 3 laptop products, added one to cart, and completed checkout at 3:15 PM. This clickstream data reveals the complete user journey.

Q2. Write a note on Session & Visitor analysis

- Session and visitor analysis are fundamental techniques in web analytics to understand user behavior:
- **Session Identification:** A session represents a sequence of activities performed by a single user during one visit to a website. Sessions are typically identified using timeout thresholds (e.g., 30 minutes of inactivity) or by detecting significant changes in user behavior.
- **Visitor Identification:** Distinguishes unique users accessing the website through cookies, IP addresses, login credentials, or device fingerprinting. Helps differentiate between new and returning visitors.
- **Session Metrics:** Key measurements include session duration, pages per session, bounce rate, entry and exit pages, and conversion rates. These metrics reveal engagement levels and content effectiveness.
- **Visitor Segmentation:** Categorizing visitors based on behavior patterns, demographics, traffic sources, or device types enables targeted analysis and personalization strategies.
- **Path Analysis:** Tracking the sequence of pages visited during sessions to identify common navigation patterns, popular content paths, and potential usability issues.

Example: Google Analytics identifies that Visitor A had 3 sessions in one week: Session 1 lasted 5 minutes with 4 page views, Session 2 lasted 12 minutes with 8 pages (made a purchase), and Session 3 was a quick 1-minute bounce. This pattern shows a returning customer who researched before buying.

Q3. Explain the difference between Influence and Homophily?

- Influence and Homophily are two key concepts in social network analysis that explain why connected individuals exhibit similar behaviors:
- **Influence (Social Influence):** The phenomenon where individuals change their behavior, opinions, or preferences due to interactions with their social connections. One person's actions directly cause changes in another person's behavior.
- **Homophily (Birds of a Feather):** The tendency for similar individuals to form connections with each other. People naturally associate with others who share similar characteristics, interests, or behaviors, rather than influencing each other post-connection.
- **Causation Direction:** Influence implies causation (connection → behavior change), while homophily represents selection (similarity → connection formation).
- **Temporal Aspect:** Influence occurs after connections are established, whereas homophily drives the initial formation of connections based on pre-existing similarities.

Key Differences:

Aspect	Influence	Homophily
Definition	Behavior change caused by social connections	Similar people form connections
Direction	Connection causes similarity	Similarity causes connection
Timing	Occurs after connection	Occurs before connection
Mechanism	Peer pressure, persuasion	Selection, attraction
Example	Friend recommends a product, you buy it	You befriend someone who likes same music

Example: On Twitter, if Alice follows Bob and then starts tweeting about cryptocurrency because Bob tweets about it, that's influence. However, if Alice followed Bob specifically because he already tweets about cryptocurrency (which she was already interested in), that's homophily.

Q4. How does social data mining work?

- Social data mining extracts meaningful patterns from social media and network data through systematic processes:

- **Data Collection:** Gathering data from social platforms using APIs, web scraping, or streaming interfaces. This includes posts, comments, likes, shares, follower relationships, and user profiles.
- **Data Preprocessing:** Cleaning and transforming raw social data by removing noise, handling missing values, normalizing text, removing duplicates, and structuring unstructured content for analysis.
- **Network Construction:** Building social network graphs where nodes represent users/entities and edges represent relationships (follower, friend, mention, retweet). This enables structural analysis of connections.
- **Pattern Discovery:** Applying data mining algorithms to identify communities, influential users, trending topics, sentiment patterns, and behavioral trends within the social network.
- **Analysis Techniques:** Utilizing graph analytics (centrality measures, clustering), text mining (topic modeling, sentiment analysis), and machine learning (classification, prediction) to extract insights.
- **Visualization and Interpretation:** Presenting findings through network graphs, word clouds, sentiment trends, and dashboards to make complex patterns understandable and actionable.

Example: A company mining Twitter data collects tweets mentioning their brand, preprocesses text by removing URLs and special characters, builds a retweet network, applies sentiment analysis to classify opinions, and discovers that negative sentiment spikes correlate with product launch delays, enabling proactive customer service.

Q5. Define opinion mining. Explain problems of opinion mining

Definition: Opinion mining, also known as sentiment analysis, is the computational study of opinions, sentiments, emotions, and attitudes expressed in text. It involves identifying and extracting subjective information from source materials to determine whether the writer's attitude toward a topic is positive, negative, or neutral.

Problems and Challenges in Opinion Mining:

- **Subjectivity and Ambiguity:** Human language is inherently subjective and context-dependent. The same words can express different sentiments in different contexts, making accurate interpretation difficult.
- **Sarcasm and Irony Detection:** Identifying when users mean the opposite of what they literally say is extremely challenging. "Great job!" could be genuine praise or heavy sarcasm.
- **Domain Dependency:** Sentiment lexicons and models trained in one domain often fail in others. "Unpredictable" is negative for cars but positive for movie plots.
- **Aspect-Level Analysis:** Users may express different sentiments about different aspects of the same product. A restaurant review might praise food quality but criticize service.

- **Negation Handling:** Words like "not," "never," or "hardly" reverse sentiment polarity, and their scope must be correctly identified. "Not bad" is actually positive.
- **Language Variability:** Slang, abbreviations, emojis, misspellings, and multilingual text in social media make preprocessing and analysis complex.

Example: A review stating "This phone has an amazing camera but terrible battery life" expresses both positive and negative opinions. A simple overall sentiment classifier would struggle, while aspect-based opinion mining would correctly identify positive sentiment for "camera" and negative for "battery life."

Q6. Short note on social network graph

- A social network graph is a mathematical representation of social relationships and interactions:
- **Structure:** Composed of nodes (vertices) representing individuals, organizations, or entities, and edges (links) representing relationships such as friendship, follows, communication, or collaboration.
- **Graph Types:** Can be directed (Twitter follows, where A follows B doesn't mean B follows A) or undirected (Facebook friendship, which is mutual). Weighted graphs assign importance values to edges.
- **Key Metrics:** Degree centrality (number of connections), betweenness centrality (bridge nodes), closeness centrality (average distance to others), and clustering coefficient (network density) measure node importance and network structure.
- **Community Detection:** Algorithms identify clusters or groups of densely connected nodes with sparse connections between groups, revealing natural communities within the network.
- **Applications:** Used for influence analysis, recommendation systems, information diffusion modeling, fraud detection, and understanding organizational structures.

Example: In a LinkedIn professional network graph, nodes represent professionals and edges represent connections. A recruiter node with high degree centrality has many connections, while someone with high betweenness centrality connects different professional communities (e.g., linking tech and finance sectors).

Q7. Challenges of social media mining

- Social media mining faces numerous complex challenges:
- **Data Volume and Velocity:** Social media generates massive amounts of data continuously. Processing millions of tweets, posts, and interactions in real-time requires scalable infrastructure and efficient algorithms.

- **Data Quality and Noise:** Social media contains spam, bots, fake accounts, irrelevant content, and unreliable information, requiring sophisticated filtering and validation mechanisms.
- **Privacy and Ethical Concerns:** Balancing data mining objectives with user privacy rights, consent requirements, and ethical use of personal information presents legal and moral challenges.
- **Multimodality:** Social media combines text, images, videos, audio, and links. Analyzing these diverse data types together requires integrated multimodal analysis techniques.
- **Dynamic Nature:** Trends, language patterns, and user behavior evolve rapidly. Models and algorithms must continuously adapt to remain effective and accurate.
- **Platform Heterogeneity:** Different social platforms have unique structures, APIs, rate limits, and data formats, making unified analysis across platforms technically challenging.

Example: During the COVID-19 pandemic, mining Twitter for public health insights faced challenges: millions of tweets daily (volume), misinformation and conspiracy theories (quality), privacy concerns about location data (ethics), and rapidly changing terminology and hashtags (dynamic nature).

Q8. Short note on evaluating recommendation

- Evaluating recommendation systems ensures they provide relevant, accurate, and useful suggestions:
- **Accuracy Metrics:** Measure how well predictions match actual user preferences using metrics like Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and precision-recall for rating predictions.
- **Classification Metrics:** For binary recommendations (relevant/not relevant), precision measures the proportion of relevant recommendations, while recall measures the proportion of relevant items successfully recommended.
- **Ranking Metrics:** Evaluate the quality of ranked recommendation lists using metrics like Mean Average Precision (MAP), Normalized Discounted Cumulative Gain (NDCG), and Mean Reciprocal Rank (MRR).
- **Coverage and Diversity:** Assess what percentage of items can be recommended (catalog coverage) and how diverse the recommendations are, avoiding filter bubbles and echo chambers.
- **User Satisfaction:** Beyond algorithmic metrics, measure actual user engagement through click-through rates, conversion rates, and explicit user feedback ratings.

Example: Netflix evaluates its recommendation system by measuring both RMSE (do predicted ratings match actual ratings?) and engagement metrics (do users actually watch recommended content?). A system with 0.9 RMSE but only 10% click-through rate is less valuable than one with 1.1 RMSE and 30% click-through.

LONG QUESTIONS

Q1. List different classical recommendation algorithms & explain any one of them

- Classical recommendation algorithms form the foundation of modern recommendation systems:
- **Collaborative Filtering (User-based):** Recommends items by finding similar users based on their rating patterns and suggesting items that similar users liked.
- **Collaborative Filtering (Item-based):** Identifies similar items based on user rating patterns and recommends items similar to those the user has already liked.
- **Content-Based Filtering:** Recommends items similar to those a user has liked in the past by analyzing item features and user profiles.
- **Matrix Factorization:** Decomposes the user-item rating matrix into lower-dimensional latent factor matrices to discover hidden patterns and make predictions.
- **Association Rule Mining:** Discovers frequent item combinations and generates "customers who bought X also bought Y" type recommendations.
- **Hybrid Methods:** Combines multiple recommendation techniques to leverage their strengths and overcome individual limitations.
- **Knowledge-Based Systems:** Uses explicit domain knowledge and user requirements to suggest items, particularly useful for infrequent purchases like houses or cars.

Detailed Explanation: Item-Based Collaborative Filtering

Item-based collaborative filtering is one of the most widely used recommendation algorithms due to its scalability and effectiveness.

- **Core Principle:** Instead of finding similar users, this algorithm finds similar items. If a user liked item A, and item B is similar to A (based on how other users rated both), then recommend item B.
- **Similarity Computation:** Calculate similarity between items using cosine similarity, Pearson correlation, or adjusted cosine similarity. Items are represented as vectors of user ratings.
- **Prediction Formula:** Predict user u 's rating for item i by taking a weighted average of u 's ratings for similar items, where weights are the similarity scores.
- **Advantages:** More stable than user-based (items change less than user preferences), better scalability (pre-compute item similarities), and provides better explanations ("recommended because you liked X").
- **Algorithm Steps:** 1) Build item-item similarity matrix, 2) For each user, identify items they've rated, 3) Find similar items to those, 4) Calculate predicted ratings using weighted sum, 5) Recommend top-N items with highest predictions.
- **Cold Start Problem:** New items with no ratings cannot be recommended until they receive sufficient ratings, requiring hybrid approaches or content-based bootstrapping.

Example: Amazon uses item-based collaborative filtering. If you rated "The Lord of the Rings" 5 stars and "The Hobbit" 5 stars, and these books have 0.95 similarity with "The Silmarillion" (because users who rated LOTR and Hobbit highly also rated Silmarillion highly), the system predicts you'll rate Silmarillion around 4.75 stars and recommends it.

Q2. How do we evaluate recommenders? Explain with accuracy metrics

- Evaluating recommendation systems is crucial to ensure they provide valuable suggestions and improve user experience:
- **Evaluation Paradigms:** Offline evaluation uses historical data and cross-validation, online evaluation tests with real users through A/B testing, and user studies gather qualitative feedback.
- **Rating Prediction Accuracy:** When systems predict numerical ratings, accuracy metrics measure how close predictions are to actual user ratings.
- **Ranking Quality:** For top-N recommendations, ranking metrics evaluate how well the system orders items by relevance rather than absolute rating values.
- **Coverage Metrics:** Measure what percentage of items and users can receive recommendations, ensuring the system isn't limited to popular items or active users.

Key Accuracy Metrics:

- **Mean Absolute Error (MAE):** Calculates the average absolute difference between predicted and actual ratings. Formula: $MAE = (1/n) \times \sum |\text{predicted} - \text{actual}|$. Lower values indicate better accuracy. Easy to interpret but treats all errors equally.
- **Root Mean Square Error (RMSE):** Similar to MAE but squares errors before averaging, penalizing larger errors more heavily. Formula: $RMSE = \sqrt{[(1/n) \times \sum (\text{predicted} - \text{actual})^2]}$. More sensitive to outliers than MAE.
- **Precision:** Measures the proportion of recommended items that are relevant. Formula: $\text{Precision} = (\text{Relevant items recommended}) / (\text{Total items recommended})$. High precision means few irrelevant recommendations.
- **Recall:** Measures the proportion of relevant items that were successfully recommended. Formula: $\text{Recall} = (\text{Relevant items recommended}) / (\text{Total relevant items})$. High recall means few relevant items were missed.
- **F1-Score:** Harmonic mean of precision and recall, providing a single metric that balances both. Formula: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$. Useful when you need a single accuracy measure.
- **Mean Average Precision (MAP):** Evaluates ranking quality by averaging precision at each position where a relevant item appears. Rewards systems that rank relevant items higher.
- **Normalized Discounted Cumulative Gain (NDCG):** Considers both relevance and ranking position, with items ranked higher contributing more to the score. Normalized to allow comparison across different recommendation lists.

- **Hit Rate:** For top-N recommendations, measures the percentage of users for whom at least one recommended item was relevant. Simple but effective for binary relevance.

Example: A movie recommender predicts you'll rate "Inception" 4.5 stars (you actually rated it 5) and "The Matrix" 3.8 stars (actual: 2). $MAE = |4.5-5| + |3.8-2| / 2 = 1.15$. $RMSE = \sqrt{[(0.5^2 + 1.8^2)/2]} = 1.33$. The RMSE is higher because it penalizes the large 1.8 error more heavily. If it recommended 10 movies and 7 were relevant: Precision = $7/10 = 0.70$.

Q3. Explain Corpus based opinion Lexicon Expansion

- Corpus-based lexicon expansion automatically grows sentiment lexicons from domain-specific text data:
- **Motivation:** General-purpose sentiment lexicons often miss domain-specific opinion words and phrases. "Unpredictable" is negative for cars but positive for mystery novels. Corpus-based methods adapt lexicons to specific domains.
- **Seed Words Approach:** Starts with a small set of known positive and negative words (seed lexicon), then iteratively expands by finding words with similar contextual usage patterns in the target corpus.
- **Contextual Similarity:** Words appearing in similar contexts likely have similar sentiment orientation. If "excellent" and "outstanding" frequently appear in similar sentence structures, and "excellent" is positive, then "outstanding" is likely positive.
- **Co-occurrence Patterns:** Analyzes word co-occurrence within sentences or windows. Words frequently co-occurring with known positive words are likely positive, and vice versa for negative words.
- **Conjunctions and Discourse Relations:** Exploits patterns like "AND" connecting same-polarity words ("good and excellent") and "BUT" connecting opposite-polarity words ("cheap but unreliable").
- **Statistical Association Measures:** Uses metrics like Pointwise Mutual Information (PMI) to measure association strength between candidate words and seed words. High PMI with positive seeds indicates positive sentiment.
- **Word Embeddings:** Modern approaches use word2vec, GloVe, or BERT embeddings to find semantically similar words in vector space. Words close to known sentiment words inherit similar sentiment.
- **Iterative Expansion:** The process repeats: newly identified sentiment words join the seed set, enabling discovery of additional words. Confidence scores help rank candidates and filter noise.
- **Label Propagation:** Constructs a graph where nodes are words and edges represent contextual or semantic relationships. Sentiment labels propagate from seed words through the graph to unlabeled words.
- **Domain Adaptation:** The expanded lexicon becomes domain-specific, capturing terminology unique to the corpus. Medical reviews might add "invasive" as negative, while tech reviews add "intuitive" as positive.

- **Validation and Filtering:** Manual review or automatic validation (checking consistency across documents) removes incorrect additions and ensures quality of the expanded lexicon.

Example: Starting with seed words {positive: [good, excellent], negative: [bad, poor]} in a laptop review corpus, the algorithm finds "lightweight" frequently co-occurs with "excellent" and "love", while "overheats" co-occurs with "bad" and "disappointed". Through PMI calculation and iterative expansion, it adds "lightweight" (positive), "overheats" (negative), "responsive" (positive), and "sluggish" (negative) to create a laptop-specific sentiment lexicon.

Q4. Explain group spam detection in detail

- Group spam detection identifies coordinated groups of users working together to manipulate reviews, ratings, or opinions:
- **Definition of Group Spam:** Unlike individual spammers, group spam involves multiple accounts collaborating to promote products, attack competitors, or artificially boost ratings. Groups are more sophisticated and harder to detect than isolated spammers.
- **Motivation for Detection:** Group spam severely damages review system credibility and misleads consumers. Detecting coordinated manipulation is critical for maintaining platform trustworthiness and fairness.
- **Behavioral Patterns:** Group members exhibit suspicious patterns: reviewing same sets of products within short timeframes, similar rating distributions, coordinated timing of reviews, and unusual similarity in review text or style.
- **Graph-Based Approaches:** Construct reviewer-product bipartite graphs or reviewer-reviewer similarity graphs. Dense subgraphs or cliques indicate potential spam groups where members heavily review same products.
- **Frequent Itemset Mining:** Apply algorithms to find frequent reviewer groups targeting the same products. Groups reviewing identical or highly overlapping product sets more often than random chance are suspicious.
- **Temporal Correlation:** Analyze review timestamps to detect synchronized activities. Group members often review target products within narrow time windows (hours or days) to maximize impact.
- **Content Similarity Analysis:** Compare review texts using cosine similarity, Jaccard similarity, or plagiarism detection. Groups may use templates, copy each other, or exhibit unnaturally similar writing patterns.
- **Rating Deviation Indicators:** Groups giving extreme ratings (all 5-star or all 1-star) that deviate significantly from other reviewers' average ratings signal coordinated manipulation attempts.
- **Account Relationship Analysis:** Investigate connections between accounts including IP addresses, device fingerprints, payment methods, or social network links. Shared infrastructure suggests coordinated accounts.

- **Statistical Outlier Detection:** Use statistical tests to identify groups whose collective behavior is statistically unlikely under normal conditions. Chi-square tests, z-scores, and anomaly detection algorithms flag suspicious groups.
- **Machine Learning Classification:** Train supervised models using features like group cohesion, review timing variance, rating patterns, and content similarity to classify candidate groups as spam or legitimate.

Example: Amazon detects a group of 15 reviewers who all posted 5-star reviews for the same obscure phone accessory within 48 hours, despite having no prior review history together. Graph analysis reveals they form a tight cluster (all reviewed the same 10 products in the same week), temporal analysis shows synchronized timing, and content analysis finds their reviews share 75% textual similarity. This evidence triggers spam group flagging and review removal.

Q5. Explain any one Opinion spam technique in detail

Detailed Explanation: Duplicate Detection Technique for Opinion Spam

- **Overview:** Duplicate and near-duplicate detection identifies spam reviews that are exact copies or slight variations of other reviews. Spammers often reuse content across multiple products or accounts to save effort.
- **Types of Duplicates:** Exact duplicates (identical character-by-character), near-duplicates (minor modifications like word substitutions or extra phrases), and template-based reviews (same structure with variable product names).
- **Exact Duplicate Detection:** Use hashing algorithms like MD5 or SHA-256 to generate unique fingerprints for each review. Identical reviews produce identical hashes, enabling instant detection with $O(1)$ lookup in hash tables.
- **Shingling for Near-Duplicates:** Break reviews into overlapping word sequences (shingles or n-grams). For example, 3-shingles of "great product quality" are {"great product quality", "product quality"}. Similar reviews share many shingles.
- **Jaccard Similarity:** Measure overlap between shingle sets. Formula: $\text{Jaccard}(A,B) = |A \cap B| / |A \cup B|$. Values above a threshold (e.g., 0.7) indicate near-duplicates. Efficient for pairwise comparisons.
- **MinHash and LSH:** Locality Sensitive Hashing (LSH) with MinHash signatures enables scalable near-duplicate detection across millions of reviews. MinHash creates compact signatures preserving Jaccard similarity, and LSH groups similar signatures efficiently.
- **Edit Distance Metrics:** Levenshtein distance counts minimum character edits (insertions, deletions, substitutions) needed to transform one review into another. Low edit distances indicate near-duplicates but are computationally expensive.
- **Cosine Similarity with TF-IDF:** Represent reviews as TF-IDF weighted term vectors and calculate cosine similarity. High similarity scores (>0.8) suggest duplicate or template-based content.

- **Cross-Product Detection:** Search for duplicates not just within the same product but across different products. Spammers copy-pasting reviews for multiple products create stronger spam signals.
- **Temporal Patterns:** Duplicates posted by the same user at different times or by different users simultaneously raise suspicion. Analyzing posting timestamps enhances detection accuracy.
- **Handling Legitimate Duplicates:** Users may legitimately repost reviews or similar products may naturally receive similar feedback. Context analysis and reviewer history help distinguish spam from legitimate duplicates.

Example: Yelp processes a restaurant review "Amazing food and excellent service! Highly recommend." Using 3-word shingling, it creates shingles: {"Amazing food and", "food and excellent", "and excellent service", ...}. Another review states "Amazing food and great service! Highly recommend." Jaccard similarity = 6 matching shingles / 8 total shingles = 0.75, exceeding the 0.7 threshold. The system flags them as near-duplicates, investigates the reviewers, and discovers both accounts created same day with only these two reviews total—clear spam indicators.

Q6. Explain web usage mining process in detail

- Web usage mining discovers meaningful patterns from user interaction data on websites:
- **Data Collection:** Gathering raw usage data from multiple sources: web server logs (access logs, error logs), application server logs, cookies, user registration databases, and client-side tracking scripts.
- **Data Preprocessing - Cleaning:** Remove irrelevant entries like image requests, stylesheet requests, robot/crawler traffic, and error records. Filter out incomplete or corrupted log entries that cannot be properly analyzed.
- **User Identification:** Distinguish individual users from IP addresses, cookies, user agents, and login credentials. Challenges include dynamic IPs, shared computers, and proxy servers requiring heuristic approaches.
- **Session Identification:** Group page accesses into logical sessions using timeout thresholds (30 minutes default), referrer information, or significant breaks in activity. Sessions represent single visits or coherent browsing episodes.
- **Path Completion:** Reconstruct missing page accesses due to browser caching or back-button usage. Analyze referrer chains and site structure to infer complete navigation paths.
- **Data Transformation:** Convert preprocessed data into suitable formats for pattern discovery. Create user-session matrices, transaction databases, or graph structures depending on mining objectives.
- **Pattern Discovery - Association Rules:** Apply algorithms like Apriori or FP-Growth to find frequently co-visited page sets. Discover rules like "users who view product A also view product B with 70% confidence."

- **Pattern Discovery - Sequential Patterns:** Identify common navigation sequences using algorithms like GSP or PrefixSpan. Discover that users typically follow "Home → Products → Details → Cart → Checkout" paths.
- **Pattern Discovery - Clustering:** Group users with similar browsing behaviors or pages with similar access patterns using k-means, hierarchical clustering, or DBSCAN. Enables user segmentation and personalization.
- **Pattern Discovery - Classification:** Build predictive models to classify users into predefined categories (buyer/browser, churner/loyal) using decision trees, neural networks, or SVM based on behavioral features.
- **Pattern Analysis and Visualization:** Interpret discovered patterns, filter trivial or redundant patterns, and visualize results through graphs, heatmaps, or dashboards. Transform technical findings into actionable business insights.
- **Application and Action:** Apply insights to improve website design, personalize content recommendations, optimize navigation structure, enhance marketing strategies, and improve user experience.

Example: An e-commerce site collects 1 million log entries daily. Preprocessing removes 400K image/CSS requests and 50K bot accesses. User identification groups remaining 550K requests into 75K unique users via cookies. Session identification creates 120K sessions (avg 1.6 sessions per user). Pattern discovery finds that 65% of users viewing "laptop accessories" also view "laptop bags" within the same session, leading to a new "Frequently Bought Together" recommendation widget that increases accessory sales by 23%.

Q7. Explain the supervised technique of sentiment classification

- Supervised sentiment classification uses labeled training data to build predictive models:
- **Overview:** Supervised learning treats sentiment classification as a standard machine learning problem where models learn from labeled examples (reviews with known sentiments) to predict sentiment of new, unseen texts.
- **Training Data Collection:** Gather large datasets of text documents with ground-truth sentiment labels (positive, negative, neutral). Sources include manually annotated reviews, star ratings mapped to sentiment, or crowdsourced labels.
- **Text Preprocessing:** Clean text by removing HTML tags, URLs, special characters; tokenize into words; convert to lowercase; remove stopwords; apply stemming/lemmatization; and handle negations and emoticons.
- **Feature Extraction - Bag of Words:** Represent text as frequency vectors of words ignoring grammar and order. Each unique word becomes a feature, and documents are vectors of word counts or binary presence indicators.
- **Feature Extraction - TF-IDF:** Weight words by Term Frequency-Inverse Document Frequency to emphasize distinctive words while downweighting common terms. Captures word importance better than raw counts.

- **Feature Extraction - N-grams:** Include word sequences (bigrams, trigrams) as features to capture local context. "not good" as a bigram has different sentiment than individual words "not" and "good."
- **Feature Extraction - Advanced:** Incorporate linguistic features (POS tags, dependency relations), sentiment lexicon scores, word embeddings (word2vec, GloVe), or contextual embeddings (BERT) for richer representations.
- **Classification Algorithms - Naive Bayes:** Probabilistic classifier applying Bayes theorem with independence assumption. Fast, effective baseline for text classification despite unrealistic feature independence assumption.
- **Classification Algorithms - SVM:** Support Vector Machines find optimal hyperplane separating sentiment classes in high-dimensional feature space. Effective for text due to sparse, high-dimensional nature.
- **Classification Algorithms - Deep Learning:** Neural networks (CNNs, RNNs, LSTMs, Transformers) automatically learn hierarchical feature representations from text. BERT and similar models achieve state-of-the-art performance through pre-training and fine-tuning.
- **Model Training:** Split data into training, validation, and test sets. Train models on training set, tune hyperparameters on validation set, and evaluate final performance on held-out test set.
- **Evaluation Metrics:** Assess performance using accuracy, precision, recall, F1-score, and confusion matrices. Evaluate across different sentiment classes separately to identify class-specific strengths and weaknesses.
- **Handling Class Imbalance:** Address skewed class distributions through oversampling minority class, undersampling majority class, synthetic data generation (SMOTE), or class-weighted loss functions during training.

Example: A movie review sentiment classifier is trained on 10,000 IMDB reviews (5,000 positive, 5,000 negative). Reviews are preprocessed and converted to TF-IDF vectors with bigram features. A linear SVM is trained achieving 88% accuracy on a test set of 2,000 reviews. The model correctly classifies "The acting was superb and the plot engaging" as positive (0.92 confidence) but struggles with "It wasn't terrible" (classified negative despite slight positive sentiment) due to negation complexity, prompting addition of negation-handling features in the next iteration.